SQL Questions for assignment:

1.Find the 2nd highest salaried person in the finance department and also find his name?

Query:

```
SELECT
   emp_no,
   first_name,
   last_name,
   (
      SELECT salary
      FROM salaries
      WHERE salaries.emp_no = employees.emp_no
        AND to_date = '9999-01-01'
   ) AS salary
FROM employees
WHERE emp_no IN (
   SELECT emp_no
   FROM salaries
   WHERE to_date = '9999-01-01'
     AND salary = (
         SELECT DISTINCT salary
         FROM salaries
         WHERE emp_no IN (
            SELECT emp_no
            FROM dept_emp
            WHERE dept_no = (
               SELECT dept_no
               FROM departments
               WHERE dept_name = 'Finance'
            )
```

```
        )
        AND to_date = '9999-01-01'
        ORDER BY salary DESC
        LIMIT 1 OFFSET 1
    )
);
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_no	first_name	last_name	salary
109084	Jiang	Vesel	134379

2.Fetch the list of of employees who worked in Production department between 1998 and 2000?

Query:

```
SELECT
  emp_no,
  first_name,
  last_name,
  hire_date
FROM employees
WHERE emp_no IN (
  SELECT emp_no
  FROM dept_emp
  WHERE dept_no = (
    SELECT dept_no
    FROM departments
    WHERE dept_name = 'Production'
  )
  AND from_date <= '2000-12-31'
  AND to_date >= '1998-01-01'
```

)

LIMIT 50000;

Note: Limited the output rows to 50,000 as it is a huge database.

Output:

First 25 rows:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

emp_no	first_name	last_name	hire_date
10003	Parto	Bamford	1986-08-28
10004	Chirstian	Koblick	1986-12-01
10010	Duangkaew	Piveteau	1989-08-24
10018	Kazuhide	Peha	1987-04-03
10020	Mayuko	Warwick	1991-01-26
10024	Suzette	Pettey	1997-05-19
10026	Yongqiao	Berztiss	1995-03-20
10029	Otmar	Herbst	1985-11-20
10030	Elvis	Demeyer	1994-02-17
10032	Jeong	Reistad	1990-06-20
10035	Alain	Chappelet	1988-09-05
10044	Mingsen	Casley	1994-05-21
10045	Moss	Shanbho...	1989-09-02
10047	Zvonko	Nyanchama	1989-03-31
10051	Hidefumi	Caine	1992-10-15
10063	Gino	Leonhardt	1989-04-08
10069	Margareta	Bierman	1989-11-05
10081	Zhongwei	Rosen	1986-10-30
10083	Vishv	Zockler	1987-03-31
10084	Tuval	Kalloufi	1995-12-15
10085	Kenroku	Malabarba	1994-04-09
10096	Jayson	Mandell	1990-01-14
10102	Paraskevi	Luby	1994-01-26
10106	Eben	Aingworth	1990-12-19
10109	Mariusz	Prampolini	1993-06-16

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Last 25 rows:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

emp_no	first_name	last_name	hire_date
439678	Adam	Encarnacion	1986-09-29
439679	Eran	Sudbeck	1986-01-22
439682	Saniya	Katalagar...	1987-04-14
439692	Kenton	Ratnakar	1986-02-02
439693	Jeong	Kusalik	1986-03-20
439698	Lijia	Feinberg	1995-11-19
439699	Izaskun	Garnier	1992-01-27
439700	Arup	Ciolek	1985-05-27
439704	Youssef	Jording	1985-05-16
439710	Girolamo	Kadhim	1985-11-05
439726	Jenwei	Picci	1986-06-02
439733	Aleksander	Staylopatis	1997-11-11
439739	Noritoshi	Denos	1988-09-29
439743	Janalee	Ernst	1986-03-26
439749	Rimli	Jahnichen	1994-10-23
439750	Gilbert	Masaki	1988-04-30
439752	Sanjeeva	Engberts	1985-03-13
439754	Hinrich	Collette	1995-12-22
439757	Gunilla	Czap	1986-02-23
439758	Zhongwei	Rodham	1992-09-25
439770	Yishai	Beznosov	1989-07-17
439771	Siamak	Brlek	1997-02-03
439776	Sadok	Auyong	1985-04-04
439782	Tristan	Ranft	1990-07-25
439783	Gay	Kadhim	1986-12-21

employees 7

3.Who is the 3rd highest salaried person in Production department without using orderby.

Query:

```
SELECT e.emp_no,
       e.first_name,
```

```
    e.last_name,
    s.salary
FROM employees e, salaries s
WHERE e.emp_no = s.emp_no
  AND s.to_date = '9999-01-01'
  AND e.emp_no IN (
      SELECT emp_no
      FROM dept_emp
      WHERE dept_no = (
         SELECT dept_no
         FROM departments
         WHERE dept_name = 'Production'
      )
  )
  AND s.salary = (
      SELECT MAX(s1.salary)
      FROM salaries s1
      WHERE s1.to_date = '9999-01-01'
        AND s1.emp_no IN (
            SELECT emp_no
            FROM dept_emp
            WHERE dept_no = (
               SELECT dept_no
               FROM departments
               WHERE dept_name = 'Production'
            )
        )
        AND s1.salary < (
            SELECT MAX(s2.salary)
            FROM salaries s2
```

```
            WHERE s2.to_date = '9999-01-01'
             AND s2.emp_no IN (
                 SELECT emp_no
                 FROM dept_emp
                 WHERE dept_no = (
                    SELECT dept_no
                    FROM departments
                    WHERE dept_name = 'Production'
                 )
             )
             AND s2.salary < (
                 SELECT MAX(s3.salary)
                 FROM salaries s3
                 WHERE s3.to_date = '9999-01-01'
                  AND s3.emp_no IN (
                      SELECT emp_no
                      FROM dept_emp
                      WHERE dept_no = (
                         SELECT dept_no
                         FROM departments
                         WHERE dept_name = 'Production'
                      )
                  )
             )
        )
    )
LIMIT 1;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

emp_no	first_name	last_name	salary
61036	Shimshon	Nergos	129361

4.Find how many employees joined in which year in a complete single report.

Query:

```
SELECT
    YEAR(hire_date) AS join_year,
    COUNT(emp_no) AS total_employees
FROM employees
GROUP BY YEAR(hire_date)
ORDER BY join_year;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

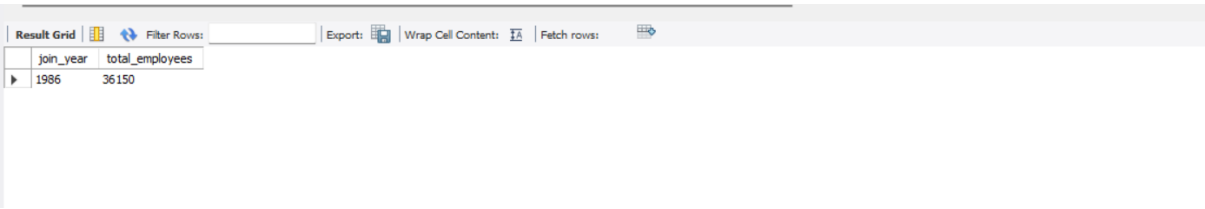
join_year	total_employees
1985	35316
1986	36150
1987	33501
1988	31436
1989	28394
1990	25610
1991	22568
1992	20402
1993	17772
1994	14835
1995	12115
1996	9574
1997	6669
1998	4155
1999	1514
2000	13

5.Fetch the maximum employees joined year from the above report without using having.

Query:

```
SELECT
    YEAR(hire_date) AS join_year,
    COUNT(emp_no) AS total_employees
FROM employees
GROUP BY YEAR(hire_date)
ORDER BY total_employees DESC
LIMIT 1;
```

Output:



join_year	total_employees
1986	36150

6.Need the reports total cost divided yearly in a table of all departments.

Query:

```
SELECT
   (
      SELECT dept_name
      FROM departments d
      WHERE d.dept_no = de.dept_no
   ) AS dept_name,
   YEAR(s.from_date) AS salary_year,
   SUM(s.salary) AS total_cost
FROM salaries s,
    dept_emp de
WHERE s.emp_no = de.emp_no
  AND s.from_date <= de.to_date
GROUP BY de.dept_no, YEAR(s.from_date)
ORDER BY dept_name, salary_year;
```

Output:

First 23 rows:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_name	salary_year	total_cost
Research	1999	368266096
Research	2000	368491055
Research	2001	364689493
Research	2002	216134039
Sales	1985	75491118
Sales	1986	161829052
Sales	1987	243817439
Sales	1988	326199268
Sales	1989	408376450
Sales	1990	494623276
Sales	1991	576404051
Sales	1992	666934396
Sales	1993	748424689
Sales	1994	829669627
Sales	1995	910246837
Sales	1996	989878517
Sales	1997	1059787336
Sales	1998	1133527658
Sales	1999	1200168809
Sales	2000	1190352531
Sales	2001	1167370083
Sales	2002	673675479

Last 23 rows:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

dept_name	salary_year	total_cost
Research	1999	368266096
Research	2000	368491055
Research	2001	364689493
Research	2002	216134039
Sales	1985	75491118
Sales	1986	161829052
Sales	1987	243817439
Sales	1988	326199268
Sales	1989	408376450
Sales	1990	494623276
Sales	1991	576404051
Sales	1992	666934396
Sales	1993	748424689
Sales	1994	829669627
Sales	1995	910246837
Sales	1996	989878517
Sales	1997	1059787336
Sales	1998	1133527658
Sales	1999	1200168809
Sales	2000	1190352531
Sales	2001	1167370083
Sales	2002	673675479

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7.Find the Year where highest female employees joined.

Query:

```
SELECT
    YEAR(hire_date) AS join_year,
    COUNT(emp_no) AS female_count
FROM employees
WHERE gender = 'F'
```

```
GROUP BY YEAR(hire_date)

ORDER BY female_count DESC

LIMIT 1;
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

join_year	female_count
1986	14434

8.Hire Date of the current manager of Human Resources department.(hint: to_Date year is 9999)

Query:

```
SELECT hire_date

FROM employees

WHERE emp_no = (

   SELECT emp_no

   FROM dept_manager

   WHERE dept_no = (

      SELECT dept_no

      FROM departments

      WHERE dept_name = 'Human Resources'

   )

   AND to_date = '9999-01-01'

);
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

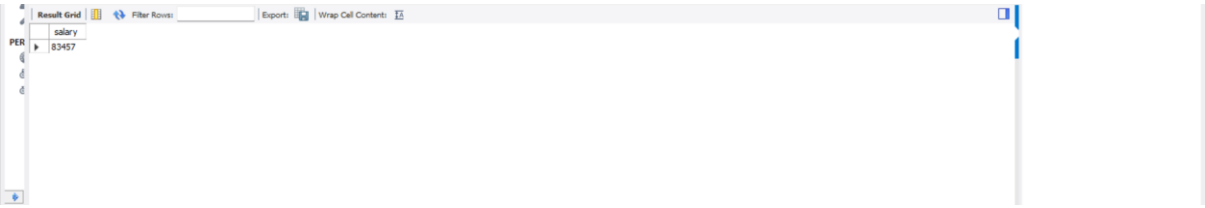
hire_date
1985-08-04

9.Salary of the current manager of the Finance dept.

Query:

```
SELECT salary
FROM salaries
WHERE emp_no = (
   SELECT emp_no
   FROM dept_manager
   WHERE dept_no = (
      SELECT dept_no
      FROM departments
      WHERE dept_name = 'Finance'
   )
   AND to_date = '9999-01-01'
)
AND to_date = '9999-01-01';
```

Output:



10.Current dept managers first names along with emp_no.

Query:

```
SELECT
   emp_no,
   first_name
FROM employees
```

```
WHERE emp_no IN (
    SELECT emp_no
    FROM dept_manager
    WHERE to_date = '9999-01-01'
);
```

Output:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_no	first_name
110039	Vishwani
110114	Isamu
110228	Karsten
110420	Oscar
110567	Leon
110854	Dung
111133	Hauke
111534	Hilary
111939	Yuchang